MTH 200
Fall 2020

Name ________________________

# Exam 1

Chapter 2 & Sections 3.1 & 3.2

**Directions:**

- Be sure to show *all* work for each question.  Solutions without supporting work will be worth little or no credit.
- Partial credit can be earned for work that makes progress towards the correct solution.
- Problems on this exam should be completed as shown in class.
- A calculator may be used on this exam.
- The exam is closed book and closed notes.
- All values should be given as exact values, unless otherwise specified.
- The exam is worth 135 points.
- You will have 90 minutes to complete & upload the exam.

| Page | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Earned | | | | | | | |
| | /36 | /20 | /7 | /18 | /27 | /27 | /135 |

1. (3pts each) Circle **true** if the given statement is always true. Otherwise circle **false**.

a) Given a function $g(x)$, if $-x^4 \le g(x) \le x^4$ for all $x$ then $\lim_{x\to 0} g(x) = 0$.

**True** or **False** ?

b) Given a function $g(x)$, if $\lim_{h\to 0} \frac{g(4+h)-g(4)}{h}$ exists, then $g(x)$ is continuous at 4.

**True** or **False** ?

c) If a function $g(x)$ is continuous at 1, then $\lim_{x\to 1} g(x) = 1$.

**True** or **False** ?

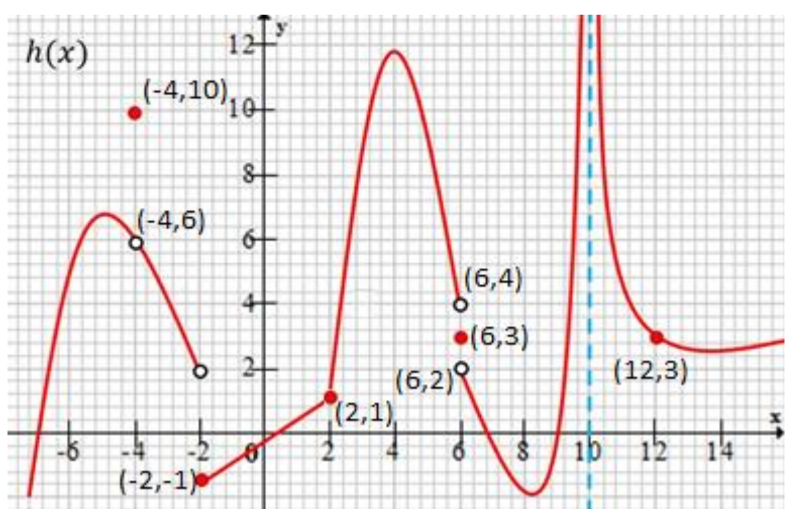
d) If a function $f(x)$ is not defined at $x = 2$, then $\lim_{x\to 2} f(x)$ does not exist.

**True** or **False** ?

e) To find when a function $f(x)$ has a horizontal asymptotes you only need to calculate $\lim_{x\to\infty} f(x)$.

**True** or **False** ?

2. (3pts each) Consider the graph of the function $h(x)$ below. Answer each of the following.




a) Find $\lim_{x\to -4} h(x)$

b) Find $\lim_{x\to -2^+} h(x)$

c) Find $h(6)$

d) Find $\lim_{x\to 10} h(x)$

e) Find $\lim_{x\to 6} h(x)$

f) State *all* the values of $x$ where $h(x)$ is <u>*not*</u> continuous.

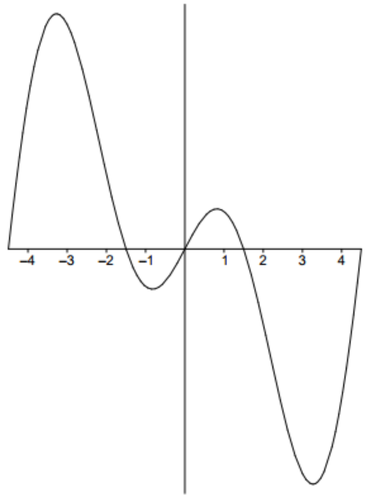
g) State *a single* value of $x$ at where $h(x)$ is continuous but not differentiable.

3. (10pts) Use the *Intermediate Value Theorem* to show that the function $f(x) = e^x - 2 + x$ has at least one real root.

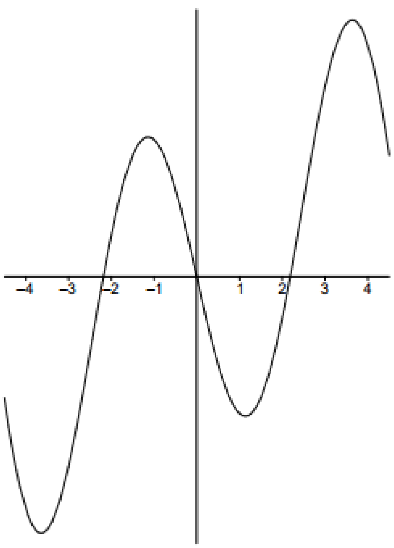
4. (10pts) What value of *c* makes the following function continuous at $x = 3$? You must use the definition of continuity to justify your answer. No credit will be given without proper justification.

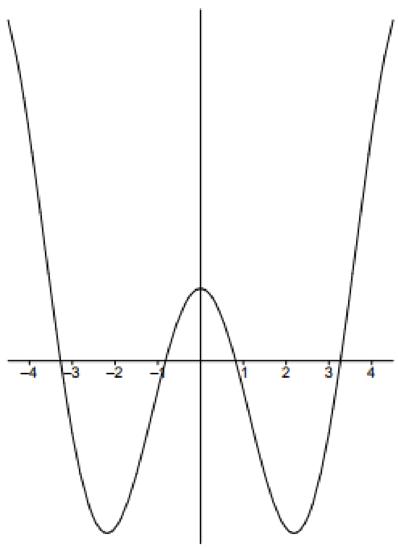
$$g(x) = \begin{cases} \frac{1}{2}x^2 + C & if \;\; x < 3 \\ 4x + \frac{1}{2} - C & if \;\; x \geq 3 \end{cases}$$

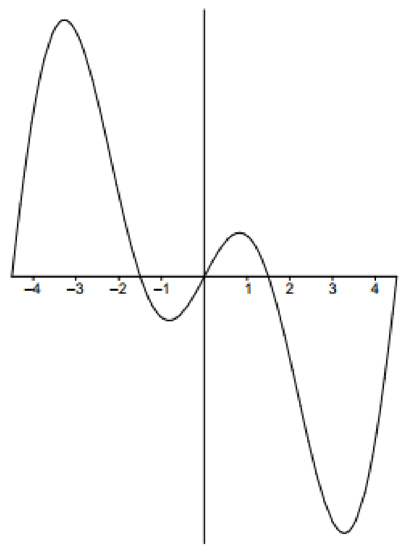
5. (7pts) The graph of the function $f(x)$ is shown below.

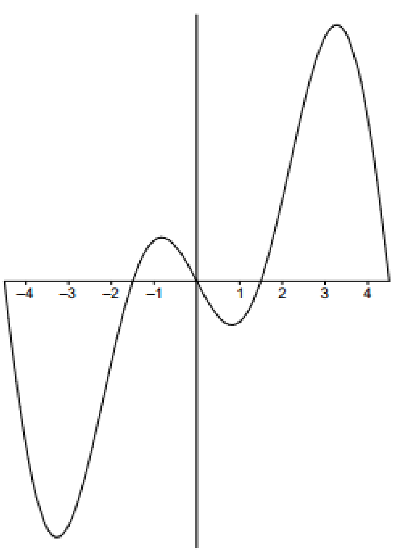


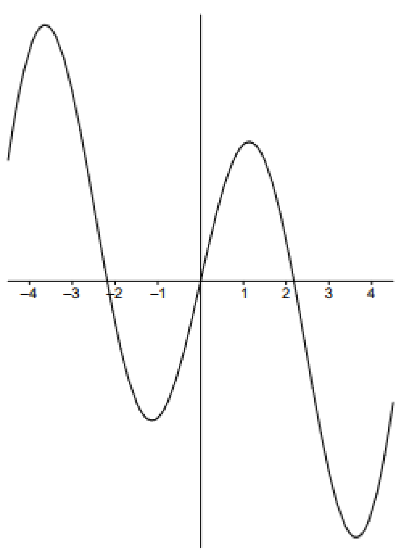

One of the six graphs below is the graph of $f'(x)$. Circle the graph of $f'(x)$.

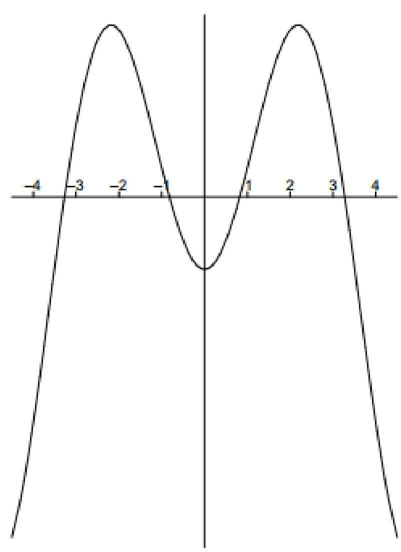


6. (10pts) Consider the function $f(x) = 2x^2 - 3x$ which has the derivative $f'(x) = 4x - 3$.
   Use the definition of a derivative as a limit to prove that $f'(x) = 4x - 3$. Show each step in your calculation and be sure to use proper notation in each step of your proof.

7. Consider again the function $f(x) = 2x^2 - 3x$ which has the derivative $f'(x) = 4x - 3$.
   a) (4pts) Find the instantaneous rate of change of the function $f$ when $x = 2$.

   b) (4pts) Find the equation of the tangent line to $f(x)$ at $x = 2$.

8. (9pts each) Evaluate the following limits *algebraically*. Show sufficient justification for each answer. An answer of *only* ‘does not exist’ is not sufficient and should be justified. For infinite limits you must state if the limit is positive or negative infinity.

a) $\lim\limits_{x\to 0} \dfrac{2}{e^x+3}$

b) $\lim\limits_{x\to 1} \dfrac{\sqrt{9x}-3}{x-1}$

c) $\lim\limits_{x\to 2^-} \dfrac{5-3x}{x-2}$

d) $\lim_{x \to -\infty} \frac{2x^2 + x}{7 - x^2}$

e) $\lim_{x \to 5} \left( \frac{10}{x^2 - 25} - \frac{1}{x - 5} \right)$

f) $\lim_{x \to \infty} \frac{-5x}{\sqrt{x^2 + 144} - 6x}$